

Mens Nova in Mundo Virtuali: Immersive Virtual Reality Experiences for Cognitive and Creative Enhancement

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Abstract

The workshop "Mens Nova in Mundo Virtuali: Immersive Virtual Reality Experiences for Cognitive and Creative Enhancement" explores the transformative potential of immersive virtual reality (VR) technologies in enhancing human cognition and creativity. By leveraging advancements in VR, brain-computer interfaces, and sensory augmentation, this workshop examines how these tools can foster neuroplasticity, improve cognitive functions, and revolutionize creative practices across disciplines. Participants will engage in discussions on topics such as adaptive VR interventions, innovative interaction paradigms, and ethical considerations in cognitive enhancement. Through interdisciplinary collaboration, the workshop aims to establish a roadmap for future research and practical applications, driving the integration of VR into everyday cognitive and creative processes.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**; *Virtual reality*; *Ubiquitous and mobile computing design and evaluation methods*; • **Applied computing** → *Psychology*; *Arts and humanities*.

Keywords

Virtual reality; Cognition; Creativity; Perception; Cognitive states; Self-enhancement; Brain-computer interfaces

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* Mens Nova in Mundo Virtuali: A New Mind in a Virtual World (in Latin)

1 Introduction

Virtual reality (VR) technologies have evolved beyond gaming and entertainment, now offering powerful tools for cognitive rehabilitation [11, 16], creative expression and thinking [10, 13], and knowledge enhancement [6, 9]. This workshop explores the intersection of immersive VR technologies with cognitive science [4] and creative applications [7, 8, 10] in the context of ubiquitous computing. We aim to bring together researchers, practitioners, and designers to examine how various types of VR experiences can be leveraged to enhance human cognitive functions and creative capabilities in everyday contexts.

Recent advances in VR technologies combined with brain-computer interfaces (BCIs), sensory augmentation, and adaptive learning systems have opened new possibilities for cognitive enhancement and creative expression. For instance, research has demonstrated that VR-based interventions can significantly improve cognitive flexibility, shifting skills, and selective attention in both clinical and non-clinical populations [1, 4]. Furthermore, fully immersive virtual reality training improved cognitive functioning in case of cognitive impairment [16].



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Similarly, immersive environments are revolutionizing creative practices across disciplines, from artistic expression to architectural design, by offering novel ways to visualize, manipulate, and experience virtual objects and spaces. For example, immersive VR design tools stimulated physical and perceptual action, leading to enhanced flexible and divergent thinking compared with two-dimensional digital design tools [12].

VR creates embodied, multisensory experiences that effectively leverage the brain's plasticity to accelerate learning and skill development. The immersive nature of VR creates a sense of presence that engages users more deeply in cognitive tasks, improving both motivation and performance outcomes.

2 Topics

Building on previous research findings, our workshop aims to explore both the theoretical foundations and practical applications of immersive VR for cognitive and creative enhancement. To these aims, this half-day workshop will address key topics, including but not limited to:

- Neuroplasticity mechanisms underlying VR-induced cognitive improvements
- Utilizing immersive environments for teaching complex concepts, to improve comprehension and retention
- Novel interaction paradigms for creative expression in immersive environments
- User experience design for cognitive workload optimization in VR
- Feedback mechanisms for immersive VR applications of cognitive and creative enhancement
- Methodologies for evaluating cognitive and creative outcomes in VR experiences
- Personalized VR interventions based on real-time cognitive state monitoring
- Ethical considerations in cognitive enhancement through immersive technologies
- Integration of VR with brain-computer interfaces for cognitive enhancement
- Social dimensions of collaborative creative processes in shared virtual spaces
- VR platforms fostering teamwork, empathy, and collective problem-solving
- Challenges in making immersive technologies inclusive for diverse populations, including individuals with disabilities
- Privacy concerns, consent, and the societal impact of using immersive technologies for cognitive augmentation

The workshop will feature a combination of paper presentations and breakout session where participants will discuss VR experiences aimed at specific cognitive or creative challenges.

2.1 Rationale of the workshop

Research demonstrates that immersive VR environments can significantly improve cognitive functions through neuroplasticity mechanisms [4], with studies showing increased gray matter volumes in motor, premotor, and supplementary motor cortices [11]. Fully immersive VR training has shown significant positive effects on global cognitive function, executive function, and attention in patients

with mild cognitive impairment [16]. VR or its combination with wearable sensors is promising tool in digital therapeutics [1, 3], digital health [5], and training [14–16].

The technology's ability to create embodied, multisensory experiences makes it particularly effective for cognitive enhancement, as it leverages the brain's plasticity to accelerate learning and skill development [1], helping with higher task accuracy [2].

Virtual reality provides explorable spaces that promote deep observation and empathy, which are crucial for design thinking and creative processes [13]. The immersive nature of VR creates a sense of presence that engages users more deeply in cognitive tasks, improving motivation and performance outcomes [1].

VR technologies have advanced beyond entertainment to aid cognitive rehabilitation, creative expression, and knowledge acquisition, making the workshop timely. The workshop brings together researchers from ubiquitous computing, cognitive science, and creative fields to create a roadmap for future interdisciplinary collaborations to maximize immersive VR's effects on cognition and creativity in everyday settings.

2.2 Objectives for the workshop

By bringing together the UbiComp/ISWC community around these emerging research directions, this workshop aims to establish a roadmap for future interdisciplinary collaborations and identify key technical and design challenges that must be addressed to realize the full potential of immersive VR for enhancing human cognition and creativity in everyday life.

We particularly encourage submissions that demonstrate novel technical approaches, empirical findings, or theoretical frameworks at the intersection of ubiquitous computing, wearable technology, and immersive VR for cognitive and creative applications.

3 Organizers

Georgi V. Georgiev is professor at the Center for Ubiquitous Computing (UBICOMP), University of Oulu, Finland. He holds a PhD in Knowledge Science from the Japan Advanced Institute of Science and Technology. His recent research focused on digital experiences enabling rehabilitation, empowerment through digital materialization, digital technology-empowered design creativity, AI in creative ideation, and neurocognition of design creativity. His works include awards for contributions to empathic virtual environments design and creativity research.

Zhengya Gong is a postdoctoral researcher at the Faculty of Art and Design, the University of Lapland, Finland. Her research focuses on topics such as Extended Reality (XR) and Human-Computer Interaction, as well as Design Creativity, Creative Methods, Design Cognition, Design Thinking, and the Influence of Culture on Design.

Iva Georgieva is an assistant professor of philosophy of science at the Bulgarian Academy of Sciences, Sofia, Bulgaria. She received her PhD in Virtual Reality narratives from the Department of History and Philosophy of Science at the University of Tokyo, Japan in 2017. She creates VR experiences with well-being goals as a Design Lead at US non-profit Studio Bahia. She has published extensively on therapeutic/interactive narrative in VR, and philosophy and ethics of medicine and technology.

Vijayakumar Nanjappan is an assistant professor in human-computer interaction at the School of Computer Science and Information Technology at the University College Cork, Ireland. His research interests focus on integrating Human-Computer Interaction, Human-Centred Design, Smart Textiles, XR, and Digital Fabrication to develop sustainable solutions that enhance user experience.

Xinhui Hu's doctoral research focuses on the intersection of Human-Computer Interaction and User-Centered Design, with a particular emphasis on empathic Virtual Reality (VR) technologies in practice. Her recent work explores how empathic VR can deepen designers' understanding of users' perspectives, allowing them to more effectively identify and address real user needs.

Amy Grech is a PhD candidate at the department of Design, Manufacturing and Engineering Management at the University of Strathclyde in Scotland. Her research explores how Virtual Reality can be applied as a tool to empower designers to experience empathy towards their users for design innovation.

Ummi Latif is a doctoral researcher in the field of Human-Computer Interaction and Virtual Reality (VR), with a particular emphasis on kinematic data analysis and visualization techniques for rehabilitation. Her research investigates how VR-based systems can enhance upper limb rehabilitation by effectively integrating detailed kinematic analysis and therapist-informed feedback. By designing VR solutions that facilitate data-driven insights for therapists and researchers, her work aims to bridge the gap between emerging technology and practical healthcare applications through user-centered and creative design methodologies.

Mengru Wang is a doctoral researcher at the Center for Ubiquitous Computing (UBICOMP), University of Oulu, Finland. She focuses on neuroscience-related tasks, including neurocognitive monitoring and behavioral analyses. She has extensive experience in neuroscience research, with a focus on design neurocognition, working memory, and meditation, using EEG and fNIRS.

Danko D. Georgiev is an assistant professor in Biochemistry at the Medical University of Varna, Bulgaria. Dr. Georgiev earned his M.D. from Medical University of Varna, in 2004, and his Ph.D. in Pharmaceutical Sciences from Kanazawa University, Japan, in 2008. His experience and research interests encompass topics of molecular neuroscience, molecular pharmacology, cognitive science, quantum chemistry, quantum physics, quantum information theory and applied computer science.

4 An estimate of expected attendance

We expect around 20–30 participants, comprising researchers, industry experts, and graduate students with an interest in immersive VR, cognitive science, and creative processes. Participation will be accessible to anyone, although it may be restricted as required to preserve an interactive and engaging atmosphere.

5 Conducting the Workshop

This half-day workshop will convene scholars, practitioners, and industry experts to examine the latest developments and obstacles in immersive virtual reality experiences aimed at cognitive and creative enhancement. Participants will partake in various presentations and productive discussions on topics including the

integration of VR with brain-computer interfaces for cognitive enhancement, the mechanisms of neuroplasticity that facilitate VR-induced cognitive improvements, innovative interaction paradigms for creative expression in immersive environments, user experience design aimed at optimizing cognitive workload in VR, and ethical considerations surrounding cognitive enhancement via immersive technologies. The workshop seeks to promote interdisciplinary cooperation, tackle practical obstacles, and investigate solutions for developing more intuitive and reliable immersive virtual reality experiences that increase cognitive and creative capabilities.

5.1 Pre-workshop actions

The appeal is sent through various UbiComp and ISWC channels and organizer networks. The organisers engage in several comprehensive research consortia and networks, aiding in the recruitment of participants for the event.

Essential information on the workshop may be found on the workshop homepage: <https://sites.google.com/view/vr-cognitive-creative-workshop>. The workshop papers are accessible, with the authors' consent, on the workshop webpage prior to the event. Workshop papers are submitted using Easy Chair.

5.2 The workshop

Every accepted workshop paper will be invited to deliver an oral presentation at the workshop. The presentations are limited to around 15 minutes to maintain a beneficial and energetic engagement. After the presentation, a coffee break is provided to facilitate participant interaction for prospective collaborations.

Participants will examine contemporary research possibilities and prospective problems in immersive virtual reality concerning cognitive and creative development and will collectively offer their ideas. The schedule is as follows.

- 9:00 - 9:15: Welcome and Introduction
- 9:15 - 10:30: Paper Presentations
- 10:30 - 11:00: Coffee Break
- 11:00 - 11:45: Breakout Session: current research opportunities and potential challenges
- 11:45 - 12:15: Group Presentation
- 12:15 - 12:30: Wrap Up

5.3 Post-workshop actions

To maintain ongoing participation and cooperation, the workshop results will be disseminated through presentation slides, discussion summaries, and recorded sessions. A participant contact list and online forum will enable networking and future partnerships. A summary of key research possibilities and problems will be provided, along with a collaborative document for future investigation. Participants are urged to develop their work into research articles and contemplate submissions to pertinent conferences or journals. Feedback will be gathered to improve future workshops, and opportunities for subsequent events, such as webinars or working groups, will be investigated to maintain momentum in immersive virtual reality research.

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A Call for papers

We welcome scholars, designers, and practitioners to submit papers to the Mens Nova in Mundo Virtuali: Immersive Virtual Reality Experiences for Cognitive and Creative Enhancement, which will investigate the potential of immersive virtual reality (IVR) in promoting cognitive and creative growth. This workshop seeks to convene professionals and rising researchers to examine existing research, identify obstacles, and offer novel paths for IVR applications across several domains, including education, design, and treatment.

The workshop aims to explore both the theoretical foundations and practical applications of immersive VR for cognitive and creative enhancement. Topic include, but are not limited to:

- Neuroplasticity mechanisms underlying VR-induced cognitive improvements
- Utilizing immersive environments for teaching complex concepts, to improve comprehension and retention
- Novel interaction paradigms for creative expression in immersive environments
- User experience design for cognitive workload optimization in VR
- Feedback mechanisms for immersive VR applications of cognitive and creative enhancement
- Methodologies for evaluating cognitive and creative outcomes in VR experiences
- Personalized VR interventions based on real-time cognitive state monitoring
- Ethical considerations in cognitive enhancement through immersive technologies
- Integration of VR with brain-computer interfaces for cognitive enhancement
- Social dimensions of collaborative creative processes in shared virtual spaces
- VR platforms fostering teamwork, empathy, and collective problem-solving
- Challenges in making immersive technologies inclusive for diverse populations, including individuals with disabilities
- Privacy concerns, consent, and the societal impact of using immersive technologies for cognitive augmentation

Important dates

July 4, 2025: Submission deadline for workshop papers
 July 19, 2025: Notification of workshop papers
 July 28, 2025: Deadline for camera-ready version of papers
 October 13, 2025: Workshop in Espoo, Finland

Workshop Organizers

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